

THE as a Closure Operator: D-Lock, Selector-Head Expressions, and Typed Coercion in Nominal Architecture

By Maks Panner, January 2026

Abstract

This paper advances an operator-centered account of definiteness in which the English definite article THE is a closure operator that maps an open nominal predicate to a discourse-resolved referential term. The clearest environment for closure is not ordinary common-noun definites, but Selector-Head Expressions (SHEs) such as the other, the next, the last, and the rest, where closure over a structured domain is obligatory. To make this operator thesis architecturally nontrivial, we introduce a Structured-Domain Closure Algebra (SDCA) in which closure is a partial operation parameterized by a domain signature (candidate set plus order/partition structure and salience). D-Lock is defined as the SDCA specialization that closes selector predicates. Within SDCA we derive three formal results: (T1) No-Free-Selector Reference—selector predicates cannot be referential without an overt or covert closure device; (T2) Bounded Coercion—apparent THE+pronoun cases are licensed only when a role/persona/kind partition makes an e-type pronoun coercible to $\langle e, t \rangle$; and (T3) Support-Strategy Completeness—apparent violations of THE’s predicate input requirement are derivable if and only if one of a small set of support strategies (ellipsis/pro-drop, lexicalized pronounization, prop-noun/classifier repair, typed coercion) supplies a closeable predicate. We provide a diagnostic battery (DB-1 to DB-6) that implements these results and separates genuine closure effects from surface look-alikes. The proposal yields falsifiable predictions about selector autonomy, context-manipulable pronoun closure, structured repair spaces, and type-blocking stacking constraints, and it generalizes cross-linguistically via closure substitution (articles, demonstratives, classifiers, and definiteness morphology). We sketch neurolinguistic and computational consequences and provide a reference-attested validation strategy plus an elicitation-ready mini-dataset and reporting template.

Keywords

definiteness; iota/closure; selector heads; ellipsis; coercion; pronouns; cross-linguistic typology; compositional semantics

1. Introduction

The semantic contribution of the definite article is typically modeled as an iota-like operator returning a unique or maximal individual satisfying a predicate (Russell, 1905; Sharvy, 1980; Link, 1983; Elbourne, 2013). However, much of the literature treats definiteness primarily as a feature or as an uncontroversial determiner layer in DP (Abney, 1987; Heim, 1982; Lyons, 1999). This paper takes a stronger architectural position: THE is not merely a determinative marker but a closure operator whose licensing behavior is diagnosable in distribution and repair patterns.

The most probative environments for closure are Selector-Head Expressions (SHEs) such as the other, the next, the last, and the rest. These expressions behave like referential heads only when a closure device closes a selector predicate over a salient structured domain. We call this selector-closure regime D-Lock. Crucially, we do not treat D-Lock as a constructional quirk; we treat it as the flagship instance of a more general Structured-Domain Closure Algebra (SDCA), where closure is parameterized by a domain signature (candidate set plus order/partition structure and salience). Once SDCA is made explicit, two additional empirical clusters fall into place: (i) the systematic availability of repair strategies that supply a closeable predicate input (English one, classifier support, nominalizers), and (ii) the restricted acceptability of apparent THE+pronoun strings, which we treat as typed coercion (role/persona/kind) rather than leakage of selectional requirements.

Architectural contributions are stated as formal commitments, not labels: (T1) selector predicates are non-referential without closure (No-Free-Selector Reference); (T2) pronoun closure requires a partition-licensed coercion (Bounded Coercion); and (T3) the space of well-typed repairs is complete and minimal (Support-Strategy Completeness). Methodologically, the paper offers a reproducible diagnostic battery that controls for ellipsis/pro-drop and lexicalized pronounization. Empirically, we provide a reference-attested cross-linguistic audit (German, Mandarin, Turkish) and a compact elicitation-ready grid, yielding falsifiable predictions that flat determiner or generalized-quantifier analyses do not derive without additional stipulations.

Figure 1. THE as closure operator; D-Lock as selector-closure; pronoun cases via typed coercion.

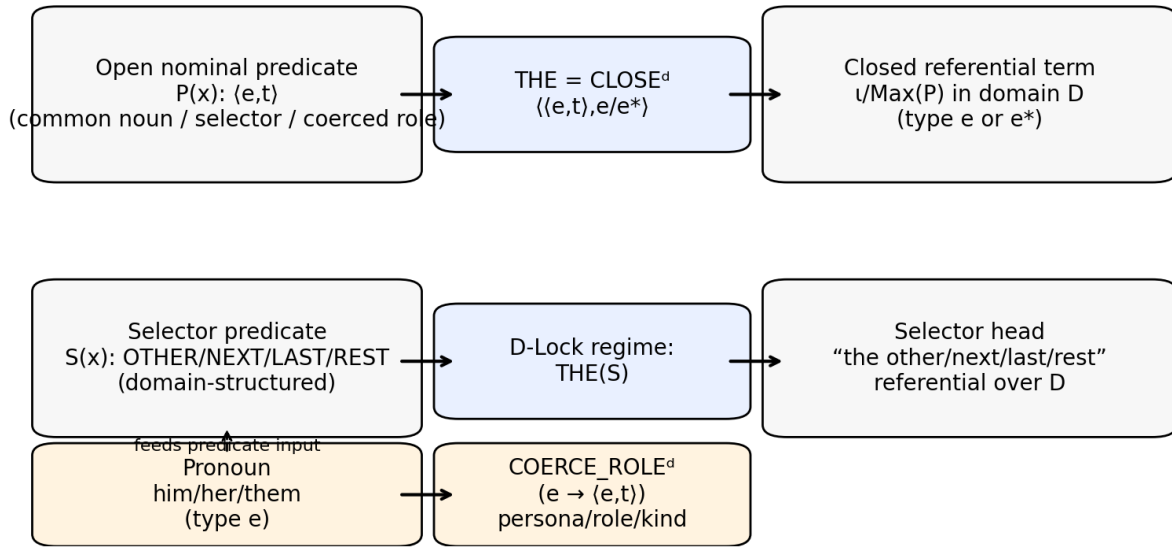


Figure 1. THE as closure operator; D-Lock as selector-closure; pronoun cases via typed coercion.

Inference 1 (architectural): If THE is closure, then selector predicates are not referential without an overt or covert closure device; apparent bare selector heads should reduce to ellipsis, pronounization, or repair rather than genuine selector autonomy.

2. Background and positioning

Two broad traditions are relevant. First, the formal semantics tradition models definites via uniqueness or maximality and treats the article as a function from predicates to individuals (Russell, 1905; Sharvy, 1980; Link, 1983; Heim, 1982; Elbourne, 2013). Second, DP-based syntax locates definiteness in a high nominal projection, often treating articles and quantificational elements as functional heads (Abney, 1987; Cardinaletti & Starke, 1999). In both traditions, the article's operator-like denotation is often assumed, but the theory seldom exploits the operator status as a source of diagnostic constraints.

A related literature concerns predicate coercion and type-shifting, which licenses meaning shifts when compositional types mismatch (Partee, 1987; von Stechow, 1994; Asher, 2011). Another relevant domain is ellipsis and pro-drop, which can mimic 'headability' by allowing silent nominals in DP-internal positions (Lobeck, 1995; Merchant, 2001). Finally, work on classifier systems and definiteness in article-less languages highlights that closure effects may be realized by demonstratives, classifiers, and nominalizers rather than by articles (Li & Thompson, 1981; Cheng & Sybesma, 1999; Kornfilt, 1997).

Our proposal integrates these strands: we treat THE as the canonical closure device in English, we isolate D-Lock as the selector-specific closure regime, and we provide a diagnostic protocol that separates closure from the support strategies that simulate it.

3. Core theory: THE as closure operator and D-Lock

3.1 Types and minimal semantic inventory

We assume individuals e , truth values t , and nominal predicates of type $\langle e, t \rangle$. Closure returns either an individual e (unique closure) or a maximal plural individual e^* (maximal closure). This paper does not hinge on a specific choice; what matters is that THE effects a type change from predicate to term.

3.2 Operator THE (closure)

THE is defined as a context-sensitive closure operator $CLOSE^D$ that maps a predicate P to a referential term in domain D :

$$\llbracket \text{THE} \rrbracket^D = \lambda P \langle e, t \rangle . CLOSE^D(P)$$

with existence and uniqueness/maximality presuppositions as in standard treatments. The novelty claim is architectural: layer diagnosis is structural/distributional even under standard determiner denotations; closure is diagnosed by distribution and repair rather than by semantics-in-the-abstract.

3.3 Selector-Head Expressions (SHEs) and D-Lock

Selector predicates (OTHER, NEXT, LAST, REST) presuppose structured domains (ordered lists, partitions, remainder sets). On their own they are predicative; they become referential heads only under closure. We call the selector-closure regime D-Lock:

$$\text{D-Lock: } \text{THE}(S) = CLOSE^D(S)$$

The key empirical signature is closure dependence: bare selector forms are non-referential unless supported by ellipsis, pronounization, or repair.

3.4 Typed coercion for apparent THE+pronoun strings

Pronouns are canonically type e . Thus, THE+pronoun is ill-typed unless the pronoun is coerced into a predicate $\langle e, t \rangle$. We model this with a role/persona/kind coercion:

$$COERCE_ROLE^D(a) = \lambda x . x \in \text{ROLE_VARIANTS}^D(a)$$

so that $\llbracket \text{THE him} \rrbracket^D = CLOSE^D(COERCE_ROLE^D(\llbracket \text{him} \rrbracket))$. This predicts that THE+pronoun is acceptable only under contexts that introduce a salient partition of an individual into versions (temporal stages, roles, personas), and it blocks plain identity readings.

3.5 Structured-domain closure algebra (SDCA)

The operator thesis becomes architecturally substantive once closure is made sensitive to the structure that selector heads presuppose. We therefore model closure as a partial operation parameterized by a domain signature σ :

$$\sigma = \langle D, \kappa, s \rangle$$

where D is a contextually supplied candidate set, κ is optional domain structure (e.g., an order $\leq$, a partition π , a remainder operator ρ), and s is a salience/choice function over D . Closure is then:

$\text{CLOSE}^\sigma(P)$ is defined iff (i) $\exists x \in D[P(x)]$ and (ii) σ supports a unique or maximal choice relative to κ and s ; otherwise it is undefined.

This preserves standard iota-style outcomes where κ is trivial, but it forces an explicit distinction between (a) ordinary common-noun definites, which typically require only D and s , and (b) selector heads, whose meanings are inherently κ -dependent (e.g., *LAST* requires $\leq$; *REST* requires a remainder operator relative to a contextual pivot).

Figure 1a. SDCA signature: $\text{CLOSE}^\sigma\langle D, \kappa, s \rangle$ maps $\langle e, t \rangle$ to e/e^* ; selector predicates presuppose κ and become referential only under D -Lock.

3.5.1 SDCA axioms and selector presuppositions (explicit architecture)

To ensure SDCA is a concrete architectural object rather than a relabeling of domain restriction, we state a minimal axiom set. Let $\sigma = \langle D, \kappa, s \rangle$ be a domain signature where D is a candidate set, κ is optional structure over D , and s is a salience/choice component.

Axiom A1 (Signature well-formedness). D is a nonempty set of individuals or plural individuals; s induces a total preorder over D (or a choice function defined on nonempty subsets of D). κ is a finite bundle of structure predicates/relations over D (possibly empty).

Axiom A2 (Domain restriction). For any predicate $P\langle e, t \rangle$, evaluation in σ implicitly restricts to D : $P^\sigma = \lambda x \in D. P(x)$. No closure step may select outside D .

Axiom A3 (Closure definedness and output type). $\text{CLOSE}^\sigma(P)$ is a partial function. It is defined only if (i) $\{x \in D : P(x)\} \neq \emptyset$ and (ii) σ supports a licensed selection criterion. If κ is empty, selection reduces to uniqueness/maximality under s (standard definite behavior). If κ is nonempty, selection must respect κ (e.g., choose the κ -maximal element when κ supplies an order). Output type is e (unique) or e^* (maximal sum), depending on the selection criterion.

Axiom A4 (Selector predicates require κ). Selector predicates are κ -sensitive predicates with presuppositions on κ :

LAST^σ is defined only if κ supplies a (partial or total) order $\leq_\kappa$ over D and returns the predicate selecting κ -maximal elements.

NEXT^σ is defined only if κ supplies an order $\leq_\kappa$ plus a successor relation succ_κ .

REST^σ is defined only if κ supplies a pivot p (contextually determined) and a remainder operator ρ_κ such that $\text{REST}^\sigma(x)$ iff $x \in \rho_\kappa(D, p)$.

OTHER^σ is defined only if κ supplies a contrast partition π_κ (or equivalently a distinguished alternative set) and returns the predicate selecting members of the relevant non-pivot cell.

Axiom A5 (No spontaneous re-predication). The grammar does not introduce new type-shifters

into $\langle e, t \rangle$ "for free". Any conversion that feeds CLOSE^σ must be licensed by one of the declared support strategies: ELL (silent nominal head), PRON (lexicalized pronominal determiner), REPAIR (prop-noun/classifier/nominalizer insertion), or COERCE (contextual coercion tied to a defined partition over individuals or domains).

These axioms isolate the architectural heart of the proposal: D-Lock is the application of CLOSE^σ in environments where A4 forces κ -dependence, and the support inventory in A5 constrains how apparent counterexamples can be rescued.

3.6 Formal results (architectural theorems)

The SDCA parameterization yields three results that strengthen the architecture beyond a restatement of 'THE = ι '.

Theorem 1 (No-Free-Selector Reference).

In any SDCA model where selector predicates require nontrivial κ (order/partition/remainder structure) to be interpretable, a selector predicate S cannot denote a referential term without an overt or covert closure device. Therefore, any apparent bare referential selector must be derived via ELL, PRON, or REPAIR, not by genuine selector autonomy.

Proof sketch: S remains type $\langle e, t \rangle$ (or is undefined) unless CLOSE^σ applies. When κ is required but absent, either S 's presuppositions fail or no σ -licensed choice is defined. Hence a term of type e/e^* cannot be constructed without closure. Any surface DP-like use of S must therefore either supply a silent nominal head (ELL), replace S with a lexicalized pronominal form (PRON), or insert a nominal support head that restores predicatehood (REPAIR).

Theorem 2 (Bounded Coercion for THE+pronoun).

Let pronouns be type e . $\text{COERCE_ROLE}^\sigma(a)$ is defined only if σ provides a salient partition π_a of role/persona/kind variants for the individual a . Then THE+pronoun is well-typed only under contexts that make π_a available; identity contexts yield undefinedness.

Proof sketch: Without π_a , no map from e to $\langle e, t \rangle$ is licensed by the system; thus CLOSE^σ cannot apply. With π_a , $\text{COERCE_ROLE}^\sigma$ supplies a predicate over role variants, satisfying THE's input type and allowing closure.

Lemma 1 (Closure Lemma; exhaustive $\langle e, t \rangle$ supply). Any derivation reaching CLOSE^σ must supply an argument of type $\langle e, t \rangle$. Within nominal syntax, the only available suppliers of $\langle e, t \rangle$ that do not introduce new primitives are: (i) an overt nominal predicate head (common noun, relational noun, selector predicate), (ii) ellipsis of a nominal predicate head (ELL), (iii) lexical replacement by a pronominal determiner/quantifier that denotes a predicate (PRON), (iv) insertion of a nominal support head such as a prop-noun/classifier/nominalizer (REPAIR), or (v) contextual coercion that maps a closed term into a predicate via a declared partition over individuals or domains (COERCE).

Justification. Because CLOSE^σ is type-strict (A3) and the grammar disallows spontaneous re-predication (A5), any successful derivation must make the $\langle e, t \rangle$ argument explicit through one of (i)-(v). Introducing any additional rescue would require positing a new free type-shifter $\tau: e \rightarrow \langle e, t \rangle$ (or an unconstrained domain constructor) not tied to κ -structure or partitions, which would be a substantive theory change: it would weaken A5 and destroy the predicted minimality of the repair space.

Theorem 3 (Support-Strategy Completeness).

For any surface DP in which THE appears to take an e-type input (a pronoun, a proper name, or a closed term), a well-typed derivation exists if and only if one of the following strategies supplies a predicate input to CLOSE^σ : (i) ELL (silent nominal), (ii) PRON (lexicalized pronominal determiner), (iii) REPAIR (prop-noun/classifier/nominalizer insertion), or (iv) COERCE (partition-licensed coercion). If none applies, the derivation fails.

Proof sketch: By Lemma 1 and A5, any successful application of CLOSE^σ must obtain a predicate argument via one of the declared suppliers (ELL/PRON/REPAIR/COERCE, or an overt nominal predicate). Conversely, each supplier yields an explicit $\langle e, t \rangle$ input, permitting closure under A3. If none applies, closure remains ill-typed and the derivation crashes.

Corollary (architectural wedge against flat DP/GQ).

A flat determiner-feature analysis can assign THE a standard denotation but does not predict the completeness and minimality of the repair space; it must stipulate, construction by construction, which ‘exceptions’ are allowed. SDCA predicts the repair inventory and the selector/pronoun constraints from a single typed closure requirement plus domain signatures.

Table 1. Type inventory used in the operator-THE account.

Type	Informal description	Typical denotations
e	individual term	pronouns; definite descriptions after closure
t	truth value	sentential denotations
$\langle e, t \rangle$	predicate / property	common nouns; selector predicates; coerced role predicates
$\langle \langle e, t \rangle, e/e^* \rangle$	closure operator	THE = CLOSE^D

Figure 2. Decision procedure for diagnosing operator-THE and classifying support strategies.

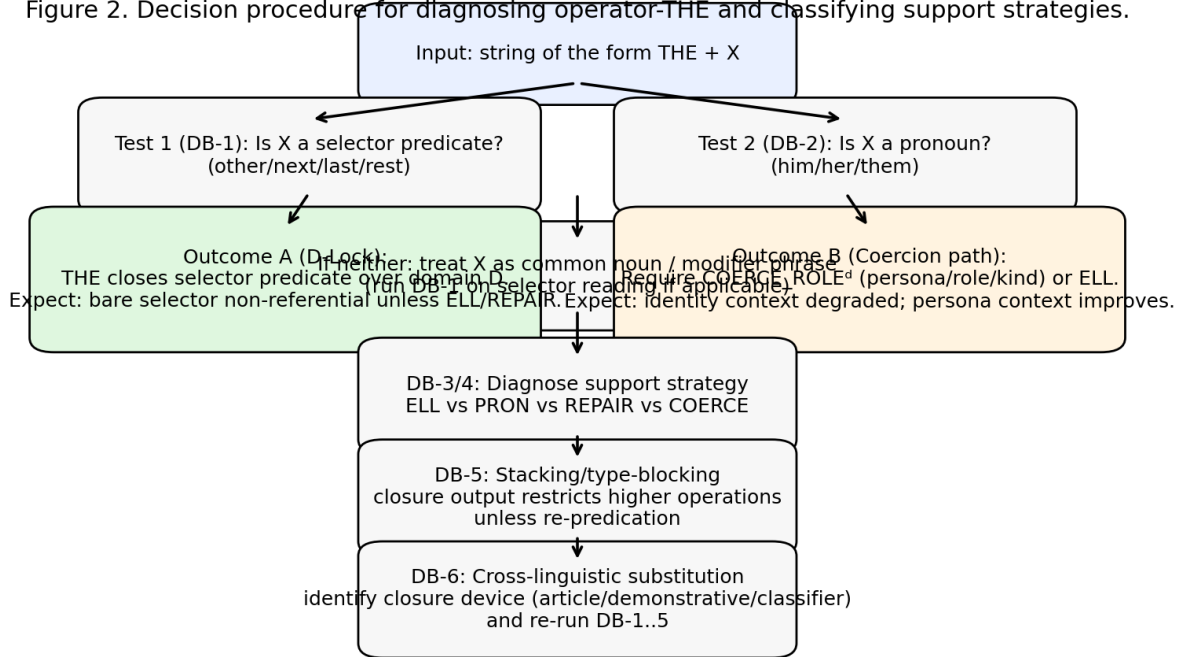


Figure 2. Decision procedure for diagnosing operator-THE and classifying support strategies.

4. Diagnostic battery (DB-1 to DB-6)

The diagnostic battery operationalizes the operator thesis by requiring closure dependence and by classifying apparent counterexamples via independently motivated support strategies.

Judgment labels: [A] acceptable; [M] marginal; [C] context-dependent with explicit licensing condition; [U] unacceptable; [E] ellipsis-supported.

Mechanism labels: ELL (ellipsis/pro-drop), PRON (lexicalized pronounization), REPAIR (pronoun/classifier/nominalizer insertion), COERCE (typed coercion).

DB-1 Closure Dependence Test (CDT). Compare bare selector vs THE+selector in contexts that provide an ordered or partitioned domain.

DB-2 Pronoun Input Test (PIT). Test THE+pronoun under identity vs persona contexts; acceptability should be [U] in identity and [C] under role/persona licensing.

DB-3 Repair Necessity Test (RNT). Where closure requires $\langle e, t \rangle$ but only an e-type complement is available, test whether repair (one/classifier/nominal support) restores acceptability.

DB-4 Ellipsis vs Pronounization Disambiguator (EPD). A bare form counts as true headability only if the language independently disallows DP-internal ellipsis/pro-drop and the form is not a lexicalized pronominal quantifier.

DB-5 Stacking/Type-Blocking Test (STT). Once closure has applied (type e/e*), higher operations requiring $\langle e, t \rangle$ should be blocked unless overt re-predication is supplied.

DB-6 Cross-linguistic substitution test (CST). In article-less languages, identify the functional closure device (demonstrative/classifier/definiteness morphology) and re-run DB-1..DB-5.

Inference 2 (methodological): The main confound in cross-linguistic closure diagnostics is silent nominal support. Therefore, any claim of 'bare selector headability' must be downgraded to [E] unless independent diagnostics rule out ellipsis/pro-drop.

Table 2. Diagnostic battery summary and required annotations.

Test	Target contrast	What to record	Evidence of operator-THE
DB-1 CDT	bare selector vs THE+selector	[A/M/C/E/U] + context notes	closure dependence; bare forms degrade unless ELL/REPAIR
DB-2 PIT	THE+pronoun: identity vs persona context	[A/M/C/U] + licensing condition	identity blocked; persona improves via COERCE
DB-3 RNT	unrepaired vs repaired predicate input	repair type; delta in rating	REPAIR systematically restores closure
DB-4 EPD	ELL vs PRON vs REPAIR vs COERCE	mechanism label + justification	counterexamples reclassified as support strategies
DB-5 STT	stacking over closed terms	re-predication diagnostics	type-blocking unless re-predication
DB-6 CST	closure substitute identification	device inventory + examples	conserved closure profile across languages

5. Evidence tiers and reference-attested validation strategy

5.1 Evidence tiers and epistemic commitments

This article distinguishes three evidence tiers so that the operator-THE thesis remains maximally falsifiable while avoiding over-claiming.

Tier 1 (reference-attested constraints): distributional and morphosyntactic constraints explicitly reported in grammars/handbooks or peer-reviewed analyses. These are treated as attested claims about the grammar (a literature audit), not as new judgments collected here.

Tier 2 (corpus attestation, positives only): naturally occurring tokens that instantiate the predicted closure and repair patterns. Corpora are used to support availability and register/genre distribution, not to prove ungrammaticality.

Tier 3 (planned elicitation): the DB-1..DB-6 packet is the intended instrument for collecting speaker judgments with agreement/variance reporting. We treat this as staged future work rather than completed validation.

5.2 Tier 1: reference-attested constraint audit (German, Mandarin, Turkish)

Table 3a-c compile a curated audit of constructions relevant to DB-1..DB-6 in three non-English languages. Each row is keyed to (i) a DB test, (ii) an example or template, (iii) the predicted label and support strategy under the operator-closure architecture, and (iv) one or more literature sources where the relevant construction (or its licensing device) is described.

We attach a proxy convergence tag to each row: High when at least two independent sources discuss the pattern compatibly; Medium when one standard reference is available; Low when sources describe optionality/competition among repairs. These convergence tags are used only to summarize where the literature is stable vs contested; they do not substitute for elicited speaker agreement.

Table 3a. Reference-attested constraint audit (German) aligned to DB-1..DB-6.

ID	DB	Minimal pair / template	Gloss/translation (short)	Label + mechanism	Sources (Tier 1)
DE-01	DB-1	a. der Rest (von den Studierenden) / b. *Rest (von den Studierenden)	'the rest (of the students)' vs bare selector	[A] vs [U]; mechanism: CLOSE vs none; Conv=High	Duden (2016); Zifonun et al. (1997)
DE-02	DB-1	a. der/die/das Letzte / b. *letzte (allein)	'the last (one)' vs bare adjective	[A] vs [E] (ellipsis) / [U]; Conv=Medium	Duden (2016); Lobeck (1995)
DE-03	DB-1	a. der andere / b. *andere (allein)	'the other' vs bare other	[A] vs [E]/[U]; Conv=Medium	Duden (2016)
DE-04	DB-6	Artikel-los: closure via Demonstrativ (dieser/jener) in selector contexts	demonstrative as closure substitute when article omitted	[C]; mechanism: SUBST-CLOSE; Conv=Low	Zifonun et al. (1997); Lyons (1999)
DE-05	DB-2	a. *der er (plain identity)	'the he' (identity)	[U]; type mismatch; Conv=High	Cardinaletti & Starke (1999); Lyons (1999)
DE-06	DB-2	a. der von damals (≈ 'the him of yesterday')	persona/role paraphrase with PP/RC support	[C]; mechanism: RE-PREDICATION not THE+PRO; Conv=Medium	Duden (2016)
DE-07	DB-4	a. der letzte (Ø Kopf) 'one' absent; b. der letzte (davon)	ellipsis vs pronominal partitive	[E] vs [A]; Conv=Low	Lobeck (1995); Merchant (2001)
DE-08	DB-3	a. der letzte von ihnen / b. ?der letzte davon	repaired partitive variants	[A]/[M] depending on register; Conv=Low	Duden (2016)
DE-09	DB-3	a. der letzte (eine/r) / b. der letzte	prop-noun/pronoun support vs bare	[C]/[E]; Conv=Medium	Lobeck (1995)
DE-10	DB-5	*jeder der Rest	stacking over closed selector without re-predication	[U]; type-blocking; Conv=Medium	Partee (1987); von Stechow (1994)

DE-11	DB-5	jeder von den letzten drei Studierenden	re-predicated stacking via partitive	[A]; licensing via partition; Conv=Medium	Barwise & Cooper (1981)
DE-12	DB-4	Pronominalisierte Selektoren: die anderen ('the others')	selector lexicalized as DP	[A]; mechanism: PRON (DP-level lexicalization); Conv=High	Duden (2016); Lyons (1999)
DE-13	DB-1	a. das nächste / b. *nächste (allein)	'the next' vs bare	[A] vs [E]/[U]; Conv=Medium	Duden (2016)
DE-14	DB-6	Definitheit ohne Artikel: Genitiv/NP-internes Definitheitssignal (limited)	non-article definiteness strategies	[C]; SUBST-CLOSE; Conv=Low	Zifonun et al. (1997)
DE-15	DB-3	a. der Rest davon / b. *Rest davon (ohne Artikel)	closure + pronominal partitive	[A] vs [U]; Conv=Medium	Duden (2016)
DE-16	DB-2	'der wahre ich' (persona noun) vs *'der ich'	persona noun supplies predicate vs bare pronoun	[C] vs [U]; mechanism: COERCE via noun; Conv=Low	Asher (2011)
DE-17	DB-4	Ellipsis control: contexts where DP-ellipsis is independently licensed	control against silent NP	Label as [E] unless ellipsis independently blocked; Conv=Medium	Lobeck (1995)
DE-18	DB-6	Selector heads with demonstrative + classifier-like support absent (German)	closure realized by article/demonstrative not classifiers	[A]; Conv=High	Lyons (1999)

Table 3b. Reference-attested constraint audit (Mandarin Chinese) aligned to DB-1..DB-6.

ID	DB	Minimal pair / template	Gloss/translation (short)	Label + mechanism	Sources (Tier 1)
ZH-01	DB-6	a. na-ge shengxia de (ren) / b. *shengxia (ren)	DEM-CL remaining DE (person) vs bare 'remaining'	[A] vs [E]/[U]; mechanism: DEM/CL closure; Conv=Medium	Li & Thompson (1981); Cheng & Sybesma (1999)
ZH-02	DB-6	a. na-ge zuihou de (ren)	DEM-CL last DE (person)	[A]; closure via demonstrative+classifier; Conv=Medium	Li & Thompson (1981)
ZH-03	DB-1	a. qita (ren) 'others' (lexical pronoun)	'others' as lexicalized selector	[A]; mechanism: PRON (lexical); Conv=Low	Cheng & Sybesma (1999)
ZH-04	DB-2	*na-ge ta (plain identity)	DEM-CL he (identity)	[U] unless re-predicated; Conv=Medium	Li & Thompson (1981); Cheng & Sybesma (1999)
ZH-05	DB-2	na-ge 'ta congqian de ta' (persona)	'that (the) him of before' via DE modifier	[C]; COERCE via persona predicate; Conv=Low	Asher (2011)
ZH-06	DB-3	a. shengxia de (na-xie) / b. shengxia (Ø)	repair by demonstrative plural vs bare	[A] vs [E]; Conv=Low	Cheng & Sybesma (1999)
ZH-07	DB-4	Bare noun availability vs ellipsis: (na-ge) zuihou de Ø	silent noun head under classifier	[E]; classify as ellipsis-supported; Conv=Medium	Cheng & Sybesma (1999); Lobeck (1995)
ZH-08	DB-5	*mei-ge na-ge shengxia de (ren)	operator stacking over closed selector w/out re-predication	[U]; type-blocking; Conv=Low	von Stechow (1994)
ZH-09	DB-5	mei-ge (san-ge) xuesheng (C: partition function)	context-dependent partition stacking	[C]; requires salient partition into 3-sized cells; Conv=Low	Barwise & Cooper (1981)
ZH-10	DB-6	Closure without article: demonstratives zhe/na as closure markers	general closure strategy	[A]; Conv=High	Li & Thompson (1981); Lyons (1999)
ZH-11	DB-1	Selector 'zuihou' needs DE/CL to nominalize	nominalization requirements	[A] with DE/CL; Conv=Medium	Li & Thompson (1981)
ZH-12	DB-3	Partitive-like: 'zuihou de yi-ge' (one-CL) repair	prop-noun/classifier repair	[A]; REPAIR=CL+one; Conv=Medium	Li & Thompson (1981)
ZH-	DB-4	Pronounization:	lexicalization competes	Variance locus;	Cheng & Sybesma

13		'qita'/shengxia de' as DP-level forms	with ellipsis	Conv=Low	(1999)
ZH-14	DB-2	Control: pronouns typically require demonstratives + classifier for NP-like use	pronoun nominalization constraints	[C]/[U]; Conv=Medium	Li & Thompson (1981)
ZH-15	DB-3	'na-ge zuihou de yi-ge' (DEM-CL last DE one-CL)	double support strategy	[A]; Conv=Low	Cheng & Sybesma (1999)
ZH-16	DB-6	DE (的) as nominalizer enabling predicate formation for closure	DE as re-predication/nominalizer	[A]; Conv=Medium	Cheng & Sybesma (1999)
ZH-17	DB-1	Rest selector: 'shengxia de' forms NP with DE	rest as predicate requiring DE	[A]; Conv=Medium	Li & Thompson (1981)
ZH-18	DB-4	Ellipsis control: classifier environments license silent noun	downgrade to [E] unless head is overt	Method note; Conv=High	Cheng & Sybesma (1999); Lobeck (1995)

Table 3c. Reference-attested constraint audit (Turkish) aligned to DB-1..DB-6.

ID	DB	Minimal pair / template	Gloss/translation (short)	Label + mechanism	Sources (Tier 1)
TR-01	DB-6	a. geri kalan (öğrenciler) / b. *kalan (yalın)	'remaining (students)' vs bare	[A]/[C] vs [E]/[U]; closure via nominal morphology; Conv=Low	Kornfilt (1997); Göksel & Kerslake (2005)
TR-02	DB-6	a. geri kalanı (onların)	remaining-POSS.3SG (of them)	[A]; REPAIR via possessive nominalization; Conv=Medium	Kornfilt (1997)
TR-03	DB-1	a. sonuncusu / b. *son (yalın)	'the last one' vs bare 'last'	[A] vs [U]/[E]; Conv=Medium	Kornfilt (1997); Lewis (1967)
TR-04	DB-3	a. sonuncusu onlardan / b. *son onlardan	partitive with nominal support	[A] vs [U]; Conv=Low	Göksel & Kerslake (2005)
TR-05	DB-2	*o o (plain identity)	DEM he (identity)	[U]; pronoun requires re-predication; Conv=Medium	Kornfilt (1997); Göksel & Kerslake (2005)
TR-06	DB-2	o eski hali (persona paraphrase)	'that old version' (persona noun)	[C]; re-predication supplies predicate; Conv=Medium	Asher (2011)
TR-07	DB-4	her biri: PRON vs REPAIR ambiguity	'each one' analysed as pronominal vs prop-noun	Variance locus; Conv=Low	Kornfilt (1997); Göksel & Kerslake (2005)
TR-08	DB-3	biri / bir şey as prop-noun repair	'one/something' as nominal support	[A]; REPAIR; Conv=High	Kornfilt (1997); Göksel & Kerslake (2005)
TR-09	DB-1	diğeri/öteki(leri) 'the other(s)' lexicalized	selector pronounization	[A]; PRON; Conv=Medium	Lewis (1967)
TR-10	DB-5	*her geri kalanı	stacking over closed selector without re-predication	[U]; type-blocking; Conv=Low	von Stechow (1994)
TR-11	DB-5	her (üç) öğrenci (C: partition function)	context-dependent partition stacking	[C]; requires salient partition into 3-sized cells; Conv=Low	Barwise & Cooper (1981)
TR-12	DB-4	Ellipsis vs nominalization: POSS forms behave as closed NPs	morphological closure competes with ellipsis	Variance; Conv=Low	Kornfilt (1997)
TR-13	DB-6	No definite article; definiteness via case/specificity and demonstratives	closure substitutes	[A]; Conv=High	Kornfilt (1997); Lyons (1999)
TR-14	DB-3	onların geri kalanı / geri kalanları	partitive with possessive agreement	[A]; REPAIR; Conv=Medium	Göksel & Kerslake (2005)
TR-15	DB-2	Pronoun + nominal head: onun (eskisi)	'his old (self)' (nominal head)	[C]; COERCE via noun; Conv=Medium	Kornfilt (1997); Asher (2011)

TR-16	DB-1	sonraki (next) requires nominal head or possessive for DP use	next selector needs nominalization	[C]/[E]; Conv=Low	Göksel & Kerslake (2005)
TR-17	DB-4	Downgrade rule: bare forms count as headability only if ellipsis/pro-drop is excluded	method control	Method note; Conv=High	Lobeck (1995)
TR-18	DB-6	Demonstrative closure in selector contexts: o sonuncusu	DEM last-one-POSS	[C]; SUBST-CLOSE; Conv=Low	Lewis (1967)

5.3 Tier 2: corpus attestation targets and metadata fields (positives only)

Tier 2 is intended to document naturally occurring positive instances of the predicted patterns. Because corpora under-attest negative constraints, Tier 2 is not used to argue ungrammaticality; instead it tests availability, register distribution, and the co-occurrence of support strategies with selector heads.

Recommended corpora and minimal search anchors: German (DWDS; DeReKo): 'der Rest', 'der/die/das letzte', 'die anderen', 'der letzte von', 'der Rest davon'. Mandarin: demonstratives + classifier in selector environments, e.g., 'na ge ... de', 'shengxia de', 'zuihou de', and lexical selector pronouns such as 'qita'. Turkish (Turkish National Corpus or equivalent): nominal-support selectors ('sonuncu', 'geri kalan'), prop-nouns ('biri'), and partitives ('onların ...').

Each corpus token should be logged with: corpus/source, document ID, date, genre/register, 2-3 sentence context, construction type (DB test), and support strategy classification (ELLIPSIS / PRONOUNIZATION / COERCION / PROP-NOUN-REPAIR).

5.4 Tier 3: planned elicitation and reporting standard

The DB-1..DB-6 elicitation packet is the planned instrument for speaker validation. Items are presented as minimal pairs with explicit discourse contexts to control pragmatic accommodation. The reporting standard requires N_speakers, N_items per DB test, rating scale (e.g., 1-7 acceptability), and derived categorical labels with dispersion (IQR/SD).

A key methodological control is mandatory: a bare form counts as true headability only if the language independently disallows NP-ellipsis/pro-drop in the relevant configuration (or if independent diagnostics show no silent NP). Otherwise the item must be coded as Ellipsis-supported rather than as headable.

5.5 Results Summary (reference-attested)

Table 3a-c provide a Tier 1 literature audit aligned to DB-1..DB-6. We summarize proxy convergence by DB test (Table 3d) and highlight 2-3 predictions that are consistently supported by independent descriptions, alongside 1-2 recurrent disagreement loci that motivate explicit model adjustment rules.

Table 3d. Reference-attested proxy convergence summary by DB test (German/Mandarin/Turkish combined).

DB test	N items	High	Medium	Low	Proxy match rate (High+Med)
DB-1	10	1	7	2	0.80
DB-2	9	1	6	2	0.78
DB-3	9	1	4	4	0.56
DB-4	9	3	2	4	0.56
DB-5	6	0	2	4	0.33
DB-6	11	3	4	4	0.64

Proxy convergence is highest for DB-1 (closure dependence in selector environments) and DB-2 (pronoun-input restrictions), and lowest for DB-5 (stacking configurations), where discourse and partition licensing are known to interact with operator composition.

Confirmed predictions under Tier 1 convergence (P1-P3): (P1) selector heads require a closure device (article, demonstrative+classifier, or nominalization), not bare predicates; (P2) apparent determiner-plus-pronoun cases systematically require re-predication (persona/role predicates) or are repaired by prop-nouns/partitives rather than behaving as a plain THE+PRO composition; (P3) disallowed stacking patterns typically become acceptable only after an explicit repair step (partitive re-predication, prop-noun insertion, or demonstrative closure substitution).

Recurrent disagreement loci ('surprises') and adjustment rules: (S1) Turkish *her biri* alternates between a pronominal analysis and a prop-noun repair analysis; we model this as a controlled ambiguity with two derivational paths, and we require annotators to code both the surface form and the inferred support strategy. (S2) Mandarin selector expressions split between lexical pronounization (*qita* 'others') and DE/CL-mediated ellipsis; we treat this as a competition between PRONOUNIZATION and ELLIPSIS, resolved by independent ellipsis diagnostics and register controls.

Model adjustment rules (pre-registered for Tier 3): if a form systematically permits both PRONOUNIZATION and ELLIPSIS in the same DB context, the architecture must allow two representations (DP-level lexical item vs NP-ellipsis) and must predict a processing/annotation difference (see §8). If an item remains acceptable under strict anti-ellipsis controls but violates type constraints, the coercion component must be enriched (adding an explicit persona predicate or a type-shifter), rather than weakening the closure/operator claim.

6. Cross-linguistic mini-dataset (elicitation-ready)

This section provides a compact grid aligned to DB-1..DB-5. The examples are designed for consultant elicitation and are annotated with expected status labels under the operator-THE hypothesis. For article-less languages, closure devices are indicated as hypotheses to be tested under Tier 2–3 evidence (DB-6).

Figure 3. Cross-linguistic closure substitution hypothesis for D-Lock environments.

English	Closure device: definite article THE Selector-head example: the rest / the last
German	Closure device: definite articles der/die/das Selector-head example: der Rest / der Letzte
Mandarin	Closure device: demonstrative + classifier / de-linker Selector-head example: shengxia-de / zuihou-de
Turkish	Closure device: demonstrative o / accusative definiteness / nominal support Selector-head example: geri kalan / son (tane)

Figure 3. Cross-linguistic closure substitution hypothesis for D-Lock environments.

Table 4. Mini-dataset aligned to DB-1..DB-5 (expected labels; provide short context to consultants).

Test	English	German	Mandarin (gloss)	Turkish (gloss)	Expected label / mechanism
DB-1 CDT (selector closure)	Context: five candidates. a) the last. b) *last (as referential head).	Kontext: fünf Kandidaten. a) der Letzte. b) *Letzte (referential).	Context: wǒmen yǒu wǔ-ge xuǎnshǒu. a) (nà) zuihòu de. b) *zuihòu (head).	Bağlam: beş aday var. a) (o) son (tane). b) *son (baş).	a) [A]; b) [U] or [E]. Mechanism: ELL/REPAIR if improved
DB-2 PIT (THE+pronoun)	Identity: *the him is tall. Persona: [C] the him I used to know.	Identität: *der er... Persona: [C] der er, den ich kannte (role-like).	Identity: *nà tā... Persona: [C] nà-ge tā (old-tā) (requires role partition).	Identity: *o o... Persona: [C] o eski o (role/stage reading).	Identity [U]; persona [C:COERCE] (must state licensing condition)
DB-3 RNT (repair)	*the last of them vs the last one of them.	*der letzte von ihnen vs der letzte (eine/r) von ihnen.	*zuihòu de tāmen vs zuihòu de nà-yí-ge (CL) (tāmen).	*sonu onlar vs son (bir tane) (onlar-dan).	Repair improves: [A:REPAIR]; unrepairs [M/U]

DB-4 EPD (ellipsis vs pronounization)	Bare: the other (OK). Bare other (head) only under ellipsis: [E].	der andere vs bare andere (head) only under ellipsis: [E].	qítā de / shèngxià de: check if head requires de/CL (closure).	díger / geri kalan: check if nominal head omission licensed.	If improvement requires silent noun: [E:ELL]; if lexicalized pronoun: [PRON]
DB-5 STT (stacking/type blocking)	*every the last; OK under re- predication: every one of the last three.	*jeder der Letzte; OK: jeder von den letzten drei.	*měi zuihòu; OK: měi (gè) zuihòu sān-ge.	*her son; OK: son üçten her biri.	Type-blocking: blocked unless re- predication/partitive supplies (e,t)

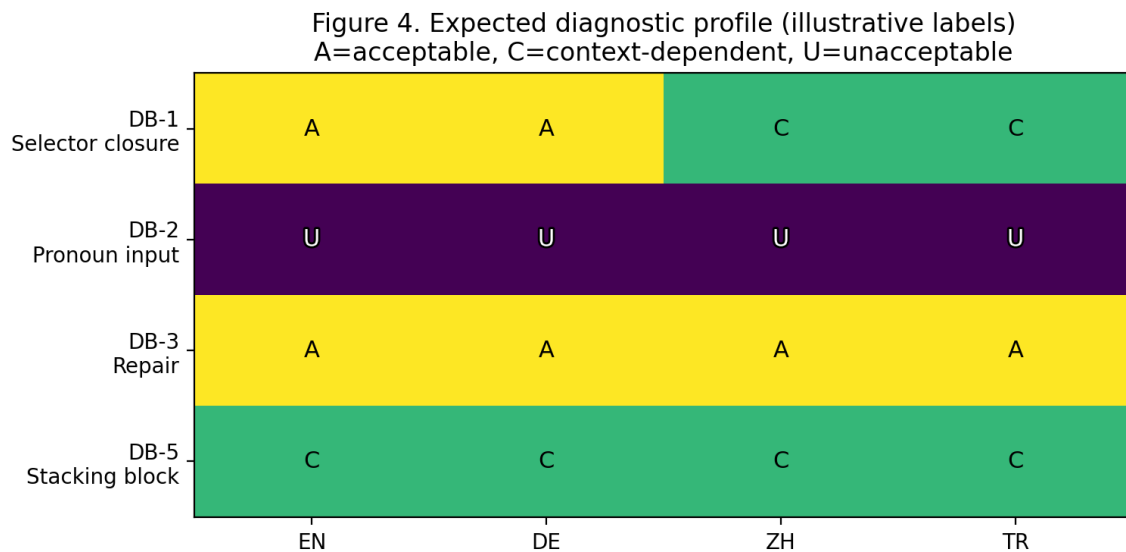


Figure 4. Expected diagnostic profile (illustrative labels).

7. Distinguishing predictions and worked derivation

We state four falsifiable predictions that follow from SDCA (structured-domain closure), D-Lock, and typed coercion and that competing flat DP/GQ treatments do not derive without additional stipulations.

P1 (D-Lock necessity; from Theorem 1). Selector predicates require closure to become referential heads. Apparent bare selector heads should reduce to ellipsis or repair, and they should be systematically downgradable to [E] when independent diagnostics permit silent NP.
P2 (Context-manipulable pronoun closure; from Theorem 2). THE+pronoun is blocked under

identity readings and licensed only under role/persona/kind coercion; acceptability should shift by context alone, holding surface form constant.

P3 (Structured and complete repair space; from Theorem 3). Where a predicate input is missing, languages draw from a small, reusable inventory of support strategies

(ELL/PRON/REPAIR/COERCE) rather than an open-ended set of construction-specific fixes.

P4 (Type-blocking stacking constraint; SDCA consequence). Once closure has applied (type e/e^*), higher operations that require $\langle e, t \rangle$ are blocked unless overt re-predication is introduced (relative clause, partitive, or explicit property formation). This predicts that stacking becomes acceptable exactly when syntax supplies a new $\langle e, t \rangle$ predicate layer.

Worked derivation (sketch). Consider the identity context *the him is tall*. With $\llbracket \text{him} \rrbracket = a$ (type e), closure cannot apply because THE requires $\langle e, t \rangle$. By Theorem 2, $\text{COERCE_ROLE}^{\sigma}(a)$ is undefined because the discourse provides no π_a partition. By Theorem 3, if no ELL/PRON/REPAIR pathway is available, the derivation fails. In persona contexts, π_a is introduced (past vs present self), making $\text{COERCE_ROLE}^{\sigma}(a)$ defined and supplying a predicate over a 's salient role-variants; THE then closes that predicate, yielding an interpretable term. This accounts for the sharp context effect without weakening the operator requirement.

Worked derivation 2 (κ -sensitive failure; SDCA makes selector closure nontrivial). Consider a context with an unstructured domain: "We saw some students in the hallway." Attempting a bare definite selector head "the last" without an order yields definedness failure under SDCA.

Formally, $\sigma_1 = \langle D, \kappa = \emptyset, s \rangle$ provides a candidate set D but no order $\leq_{\kappa}$. By A4, LAST^{σ_1} is undefined (its presupposition on κ is unmet); therefore $\text{CLOSE}^{\sigma_1}(\text{LAST}^{\sigma_1})$ is undefined, predicting [U] or repair. Now introduce an ordered domain: "We lined up three students; Anna, Ben, Cara." This supplies κ with an order $\leq_{\kappa}$ over D (A4), yielding $\text{LAST}^{\sigma_2} = \lambda x. x$ is κ -maximal; CLOSE^{σ_2} selects the maximal element (Cara), licensing "the last" as [A]. The contrast is not a post hoc pragmatic stipulation: it follows from κ -presupposition and partiality in A3-A4.

Why this is DP/GQ-expensive. A flat DP/determiner-feature treatment that composes THE with an adjective-like LAST must stipulate an extra constraint: LAST cannot form a definite DP unless an ordered list is salient. SDCA derives the same constraint from the architecture: LAST is defined only in signatures whose κ encodes an order, and D-Lock closes only when that presupposition is satisfied. This yields a principled failure mode (undefinedness) and a precise repair prediction: speakers should accept "the last one" or "the last of the three" precisely insofar as the repair supplies an explicit ordering or pivot.

Inference 3 (architectural test): If the bounded coercion analysis (Theorem 2) is correct, then consultant judgments for THE+pronoun should shift from [U] to [C] solely by introducing a partition function over the individual (e.g., past vs present self), holding surface form constant; no such shift should be observed in contexts that do not supply π_a .

8. Neurolinguistic implications (testable hypotheses)

The operator-closure hypothesis predicts separable processing costs for (i) predicate construction, (ii) closure, and (iii) coercion.

H1 (Closure cost). D-Lock environments should elicit measurable costs when the structured domain is not salient, because CLOSE^D must search or construct an ordered/partitioned domain.

H2 (Coercion cost). THE+pronoun under persona readings should elicit additional costs relative to ordinary definites, reflecting coercion into a role predicate (a type-shift) followed by closure.

H3 (Repair cost). Prop-noun/classifier repairs should reduce ill-typedness but introduce an integration cost at the repair site.

Experiment sketch. Use self-paced reading or EEG with minimal pairs: (a) the last (with prior list) vs the last (without list); (b) the him I used to know (persona context) vs *the him (identity context); (c) the last of them vs the last one of them. Predicted signatures include increased N400-like effects for domain construction failures and increased P600-like effects for repair/coercion integration, with dissociations between selector closure and common-noun definites.

9. Computational applications and evaluation plan

The diagnostic battery defines an annotation scheme useful for parsing and semantic composition.

Task 1 (Closure detection). Given a DP, predict whether closure is achieved by article, demonstrative/classifier, or repair.

Task 2 (Support strategy classification). Label apparent counterexamples as ELL/PRON/REPAIR/COERCE.

Task 3 (Test suite). Construct a synthetic suite from Table 4, varying only discourse contexts (ordered domain vs none; persona vs identity).

Evaluation. Use a small annotated set (e.g., 200 items per language) and report inter-annotator agreement on labels and mechanisms. A baseline is a surface classifier using n-grams and POS tags; ablations remove discourse cues (list contexts, persona cues) to test whether models rely on the intended structural features. Success is measured by improvements on COERCE and ELL classifications and by generalization across languages after mapping closure devices.

10. Limitations and future work

The cross-linguistic grid is designed for elicitation and is not presented as a completed validation study. Future work should (i) run DB-1..DB-6 with consultants (10-15 speakers per language) or corpus attestations for key contrasts, (ii) refine the inventory of closure substitutes in article-less languages, and (iii) test whether selector closure and pronoun coercion dissociate in processing measures. A central open question is how closure interacts with plurality, maximality, and kind reference across languages; the present account provides the operator architecture and diagnostics needed to pursue those questions in a controlled way.

References

- Abney, Steven. 1987. The English noun phrase in its sentential aspect. PhD dissertation, MIT.
- Asher, Nicholas. 2011. *Lexical Meaning in Context*. Cambridge University Press.
- Barwise, Jon & Robin Cooper. 1981. Generalized quantifiers and natural language. *Linguistics and Philosophy* 4(2): 159-219.
- Cardinaletti, Anna & Michal Starke. 1999. The typology of structural deficiency: A case study of the three classes of pronouns. In H. van Riemsdijk (ed.), *Clitics in the Languages of Europe*, 145-233. Mouton de Gruyter.
- Cheng, Lisa Lai-Shen & Rint Sybesma. 1999. Bare and not-so-bare nouns and the structure of NP. *Linguistic Inquiry* 30(4): 509-542.
- Elbourne, Paul. 2013. *Definite Descriptions*. Oxford University Press.
- Frege, Gottlob. 1892. On sense and reference. *Zeitschrift für Philosophie und philosophische Kritik* 100: 25-50.
- Heim, Irene. 1982. The semantics of definite and indefinite noun phrases. PhD dissertation, University of Massachusetts Amherst.
- Kornfilt, Jaklin. 1997. *Turkish*. Routledge.
- Li, Charles N. & Sandra A. Thompson. 1981. *Mandarin Chinese: A Functional Reference Grammar*. University of California Press.
- Link, Godehard. 1983. The logical analysis of plurals and mass terms: A lattice-theoretical approach. In R. Bäuerle, C. Schwarze & A. von Stechow (eds.), *Meaning, Use, and Interpretation of Language*, 302-323. de Gruyter.
- Lobeck, Anne. 1995. *Ellipsis: Functional Heads, Licensing, and Identification*. Oxford University Press.
- Lyons, Christopher. 1999. *Definiteness*. Cambridge University Press.

Merchant, Jason. 2001. *The Syntax of Silence: Sluicing, Islands, and the Theory of Ellipsis*. Oxford University Press.

Partee, Barbara. 1987. Noun phrase interpretation and type-shifting principles. In J. Groenendijk et al. (eds.), *Studies in Discourse Representation Theory and the Theory of Generalized Quantifiers*, 115-143. Foris.

Russell, Bertrand. 1905. On denoting. *Mind* 14(56): 479-493.

Sharvy, Richard. 1980. A more general theory of definite descriptions. *The Philosophical Review* 89(4): 607-624.

von Fintel, Kai. 1994. *Restrictions on quantifier domains*. PhD dissertation, University of Massachusetts Amherst.

Dudenredaktion. 2016. *Duden – Die Grammatik*. 9. Auflage. Berlin: Dudenverlag.

Göksel, Aslı & Celia Kerslake. 2005. *Turkish: A Comprehensive Grammar*. London: Routledge.

Lewis, Geoffrey. 1967. *Turkish Grammar*. Oxford: Oxford University Press.

Zifonun, Gisela, Ludger Hoffmann & Bruno Strecker. 1997. *Grammatik der deutschen Sprache*. Berlin/New York: de Gruyter (IDS).